Los Angeles, CA
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Royce Mettler

AI Engineer, Software Developer, Educator

Portfolio: roycemettler.com
github.com/RoyceMettler
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Experienced AI engineer with expertise in AI-driven business solutions including machine learning and generative AI. Northwestern University graduate with a double BA in Mathematics and Computer Science with a concentration in AI. Innovative thinker and risk-taker who combines technical expertise with strong communication skills and cross-functional collaboration. Approachable, solution-oriented, and friendly. Outside of tech, an avid pianist and music producer using Ableton Live.

Experience

JUNIOR AI ENGINEER **September 2023 — August 2024**
Vitu *Agoura Hills, California & Remote from Evanston, Illinois*

- Spearheaded the full-stack development of a 92% accurate internal-use neural network designed to diagnose and predict customer churn, enabling account managers to anticipate client needs and design retention strategies.
- Performed extensive data preprocessing (one-hot, feature scaling, new feature engineering) and data augmentation (SMOTE).
- Leveraged SHAP to generate per-customer heatmaps identifying the most influential features contributing to churn predictions, enabling targeted retention interventions by business stakeholders and leading to actionable insights.
- Collaborated with the interstate account management team to align the model's outputs with real-world customer behaviors, directly influencing revenue protection.

AI ENGINEERING INTERN **June 2023 — August 2023**
Vitu *Agoura Hills, California*

- Spearheaded the full-stack development of an internal-use AI agent powered by GPT-4 that answers questions about interstate vehicle registration and titling in the United States.
- Designed and implemented a frontend website using jQuery through AJAX to send and receive messages from the bot which was hosted on a Python server. Later, containerized the bot with Docker and deployed with Kubernetes.
- Collaborated with an international team of testers to iteratively optimize the chatbot's performance and troubleshoot issues.
- Live demo'd the chatbot capabilities to the c-suite resulting in them rehiring me for part-time work throughout the school year.

STUDENT AI RESEARCHER **September 2023 — June 2024**
Cognition, Creativity, and Communication (C3) Lab *Evanston, Illinois*

- Designed and implemented a proprietary model called *Satyrn* to reduce hallucinations in LLM outputs by dynamically grounding responses in real-world databases, enabling verifiably factual report generation at scale.
- Developed and deployed new report templates including time-series trend analysis, entity benchmarking, and top-entity rankings to expand Satyrn's utility across diverse datasets.
- Participated in interdisciplinary discussions on ethical AI, the societal impact of generative systems, and real-world deployment strategies across healthcare, finance, and media.

Education

Bachelor of Arts, *Northwestern University (Evanston, IL)* September 2021 — June 2025
Double Major in Mathematics and Computer Science with a Concentration in Artificial Intelligence
Study Abroad, *Yonsei University (Seoul, South Korea)* August 2024 — December 2024
Formally stepped away from Vitu to study abroad

Relevant Coursework	Data Structures and Algorithms I & II, Artificial Intelligence, Machine Learning, Deep Learning, Causal Inference, Programming Languages, Computer Systems, Computer Networks, Multi-variable Calculus I & II, Linear Algebra I & II, Real Analysis I, II, & III, Optimization, Number Theory, Topology, Statistics
Extracurricular Activities	Taught Python to 5th Graders (2023, 2024), STEM tutor, One-on-one Tech Support for Senior Citizens, Chair in Model UN, Bobarista at Möge Tee Evanston, Created YouTube Content About Northwestern

Skills

Programming Languages	Python (including OpenAI API, PyTorch, NumPy), C++, C#, Basic Typescript, Basic React
Software	Excel, Word, Powerpoint, Docker, Kubernetes, BitBucket, Github, Unity, LaTeX, Ableton
Business & Leadership	Executive Communication, Interpersonal Communication, Cross-Functional Collaboration, Strategic Planning, Organization, Very Detail-Oriented, Adaptable, Flexible Traveler
Spoken Languages	English (Native), Spanish (Professional Working Proficiency), Korean (Elementary Proficiency)

References

Ashish Khosla	Head of Platform and Technology at Vitu	akhosla@vitu.com
Joann Disney	Program Manager at Vitu	jdisney@dmvdesk.com
Kristian Hammond	Prof. of CS and Journalism At Northwestern, Head of C3 Lab	Kristian.Hammond@northwestern.edu